An impact of the diet on serum fatty acid and lipid profiles in Polish vegetarian children and children with allergy.

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BACKGROUND/OBJECTIVES: Vegetarian diet has become an increasing trend in western world and in Poland. The frequency of allergies is growing, and the effectiveness of vegetarian diet in allergic diseases is a concern for research. We aimed to study an effect of vegetarian diet on lipid profile in serum in a group of Polish children in Poland and to investigate lipid parameters in healthy vegetarian children and in omnivorous children with diagnosed atopic disease. SUBJECTS/METHODS: Serum lipid profiles (triglycerides, total cholesterol, low-density lipoprotein (LDL) and high-density lipoprotein (HDL) cholesterol, fatty acids) were assessed in groups of children: healthy vegetarians (n=24) and children with diagnosed atopic diseases (n=16), with control group of healthy omnivores (n=18). Diet classification was assessed by a questionnaire. RESULTS: No differences were observed in serum triglycerides, LDL cholesterol and saturated and monounsaturated fatty acids level in all groups. In the group of Polish vegetarian children, we recorded high consumption of vegetable oils rich in monounsaturated fatty acid, and sunflower oil containing linoleic acid. This observation was associated with higher content of linoleic acid in serum in this group. Among polyunsaturated n-6 fatty acids, linoleic acid revealed significantly ($P<0.05$) lower levels in allergy vs vegetarian groups. In case of eicosapentaenoic acid (n-3 fatty acid), the allergy group showed higher levels of this compound in comparison to vegetarians. CONCLUSIONS: Significantly higher concentration of linoleic acid in vegetarian children in comparison to allergy group indicated possible alternative path of lipid metabolism in studied groups, and in consequence, some elements of vegetarian diet may promote protection against allergy.